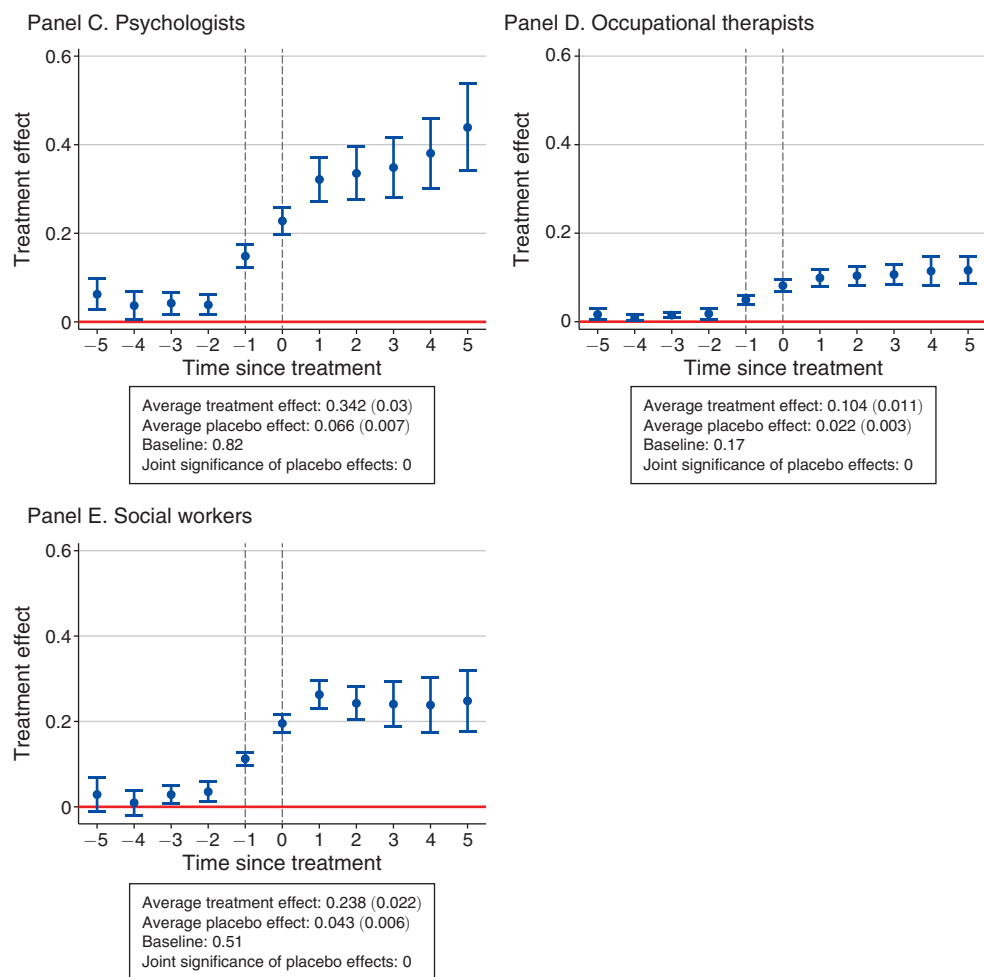


FIGURE 3. EFFECTS OF CAPS ON MENTAL HEALTH PRACTITIONERS, PRACTITIONERS PER 10,000 PEOPLE (*continued*)

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for the CAPS' effects on the number of mental health practitioners per 10,000 people. We present the effects on the overall number of mental health practitioners—the sum of the next categories (panel A), psychiatrists (panel B), psychologists (panel C), occupational therapists (panel D), and social workers (panel E). Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for the treated in the first pre-CAPS period.

the first year after CAPS' adoption. Treatment effects then rise until the fifth year, reaching 0.43 professionals per 10,000 people (53 percent, s.e., 0.05). One year after the program, the effects of CAPS on the rates of occupational therapists and

TABLE 1—EFFECTS OF CAPS ON THE SUPPLY OF MENTAL HEALTH PRACTITIONERS: NUMBER OF PROFESSIONALS PER 10,000 PEOPLE

	2-year effect (1)	5-year effect (2)	Placebo effect (3)	Mean at baseline (4)
<i>Mental health practitioners</i>				
Total	0.787 (0.051)	0.834 (0.053)	0.159 (0.014)	1.82
Psychiatrists	0.164 (0.014)	0.151 (0.015)	0.029 (0.004)	0.316
Psychologists	0.295 (0.030)	0.342 (0.030)	0.066 (0.007)	0.819
Therapists	0.095 (0.011)	0.104 (0.011)	0.022 (0.003)	0.173
Social workers	0.234 (0.021)	0.238 (0.022)	0.043 (0.006)	0.512
<i>Other types of health care practitioners</i>				
Family doctors	0.030 (0.037)	0.044 (0.036)	0.016 (0.010)	1.332
General practitioners	0.003 (0.068)	0.057 (0.071)	0.035 (0.021)	2.660
Overall supply of physicians	0.152 (0.087)	0.286 (0.096)	0.097 (0.032)	5.620

Notes: This table reports the average effects of CAPS on the number of mental health practitioners working in the public system per 10,000 people. Standard errors in parentheses are computed using a municipality-level clustered bootstrap. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. Columns 1 and 2 report the average effect after two years and after five years, respectively. Column 3 reports the placebo effect. Column 4 shows the mean of each variable at baseline—i.e., one period before the event.

social workers are 0.1 (60 percent, s.e., 0.01) and 0.26 (50 percent, s.e., 0.02), respectively. Unlike what we see for psychologists, these effects remain relatively stable in subsequent years.

Two important observations relate to the effects we see on the number of mental health professionals. First, we estimate an increase in the year before CAPS open. This increase happens because municipalities need to formally hire professionals well before the actual opening, since the bureaucratic process for hiring professionals is somewhat lengthy.¹⁴ Later, we show how the results on demand for mental care help to strengthen this point. The second observation relates to the size of the increase, which might seem suspiciously large at first. However, as mentioned in the background section, psychiatrists only work part-time at CAPS. Hence, it is easy for professionals working in the private sector to also work at CAPS and accommodate the center working hours in their schedules.

¹⁴ We have some anecdotal evidence from private conversations with municipality health officials indicating that this timing mismatch between the hiring process and the CAPS actual opening frequently happens.

The same observations made for psychiatrists are valid for other professionals. The only difference is that professionals who have to work more time at CAPS are easier to train, and the yearly supply of new graduates is significantly larger than that of psychiatrists.¹⁵

In online Appendix Table A5, we show the policy effects on overall hours worked by each type of professional. The results indicate that the policy had a positive effect on hours worked and are in line with our results on the number of professionals. Our results indicate a *de facto* expansion in the supply of mental health care, instead of just a substitution from other parts of the health care system. This increase is significantly smaller for psychiatrists, which is consistent with our prior discussion that psychiatrists work only part-time at CAPS.

Finally, since CAPS are viewed as a substitute for psychiatric hospitals, we look at the effects of CAPS on the number of psychiatric beds in online Appendix Figure A7. We find no pre-trends nor any statistically significant impact, although the imprecisely estimated effects starting three years after the program suggest a slight decrease. In any case, these effects do not appear immediately after the program like most of our results, especially those regarding mental health inpatient care (to be shown next). Therefore, if anything, the decrease in the supply of psychiatric beds is a consequence of introducing CAPS, instead of being part of the treatment itself.

Overall, our results indicate that the reform represents a large increase in the local supply of mental health providers, without a significant decrease in the supply of inpatient services. These results may be particularly important for small municipalities with a limited supply of such professionals.

B. Effects on Equilibrium Outcomes

Outpatient Care.—Figure 4 and Table 2 present the effects of CAPS on the availability of outpatient, community-based care. We find that after a CAPS opens, the number of outpatient procedures performed by mental health professionals increases remarkably: on average, they increase by 264.2 procedures per 10,000 people per year (or 113 percent of the baseline mean, s.e., 23.6). Treatment effects for the first year are 132 (s.e., 13.7) for psychiatrists, 83 (s.e., 9.6) for psychologists, 18 (s.e., 4.4) for occupational therapists, and 34 (s.e., 4.6) for social workers. Relative to the average within the treated in the pre-CAPS period, these effects are 197 percent for psychiatrists, 75 percent for psychologists, 74 percent for occupational therapists, and 94 percent for social workers. The effect on the number of outpatient procedures performed by psychiatrists decreases in the subsequent years, while it increases over time for the procedures performed by other professionals.

As previously mentioned, outpatient procedure codes change over time. Therefore, we avoid evaluating specific procedures related to mental health. There is one exception, though: occupational therapies, which involve activities like craft, music, and

¹⁵ If we look at estimates for the number of these professionals between 2002 and 2015 using the National Household Sample Survey—with the caveat that some categories of occupations are somewhat coarser than what would be ideal—all mental health providers had their numbers increased by at least 88,000 during this period, while the number of professionals needed between 2002 and 2016 based on our estimates is around 5,000.

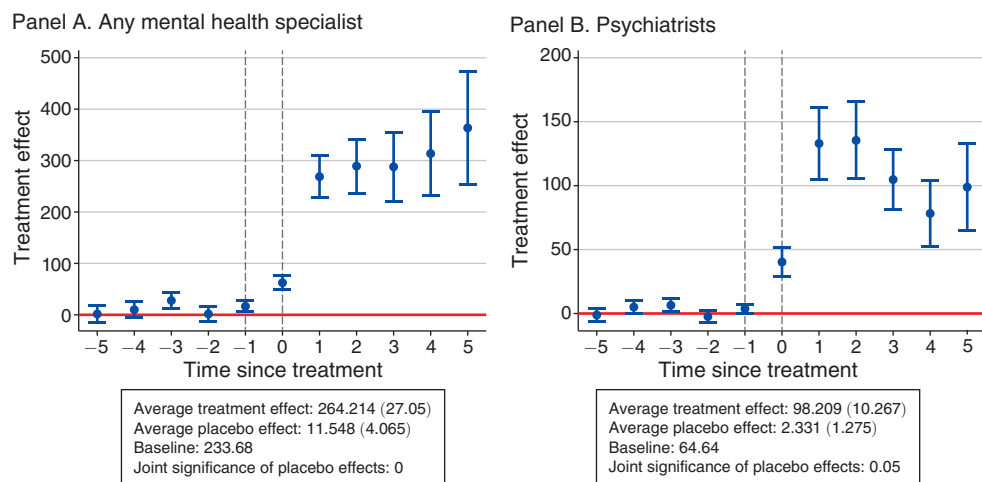


FIGURE 4. EFFECTS OF CAPS ON OUTPATIENT CARE BY MENTAL HEALTH SPECIALISTS: NUMBER OF OUTPATIENT PROCEDURES PERFORMED BY EACH TYPE OF SPECIALIST PER 10,000 PEOPLE

(continued)

dance for therapeutic purposes. In principle, this kind of procedure can be delivered at any primary-care health facility. Table 2 shows that therapeutic workshops increase by approximately 0.7 per 10,000 inhabitants after a CAPS is introduced, or a 60 percent increase, compared to the average in the pre-CAPS period.¹⁶

We also look at the effect of CAPS on the distribution of antipsychotic drugs. These are used mainly to treat schizophrenia. We find that the number of antipsychotic medications distributed (per 10,000 people) increases steadily in treated areas after the policy reform, compared to control municipalities (see Table 2 and online Appendix Figure A6). Our estimates are noisy and statistically insignificant, but their sizes are comparable to the pretreatment mean. In particular, these procedures increase from 3.7 (s.e., 6.8) two years after CAPS open, to 16.7 (s.e., 12.4) five years after (or 48 percent and 217 percent relative to the baseline mean, respectively). As medical therapy is one of the most common treatments within mental health outpatient care, this is another piece of evidence consistent with CAPS increasing utilization of outpatient mental health care.¹⁷

Inpatient Care.—Next, we investigate the effects of CAPS on mental health hospital admissions (per 10,000 people). Figure 5 and Table 3 show that the opening of a CAPS leads to a sharp reduction in mental health hospitalization rates: the average

¹⁶The event-study is shown in online Appendix Figure A6.

¹⁷We also estimate the relationship between the policy reform and psychosocial care procedures (per 10,000 people), a measure comprising all the outpatient services delivered at CAPS. Although this association should be mechanical, it provides evidence about the timing and compliance with the policy. Online Appendix Figure A5 shows that psychosocial care procedures jump right after the introduction of CAPS and then rise steadily, suggesting wide use of these centers by the local population as soon as they are implemented. As already discussed in the background section, most of the services delivered at CAPS target patients with schizophrenia (panel B).